

File Edit Selection View Go ...

Streamlit

EXPLORER

0.Image analysis

final_app.py

np_image.py

np1_image.py

code1.py

firstapp.py

myimage.py

OUTLINE

TIMELINE

Welcome

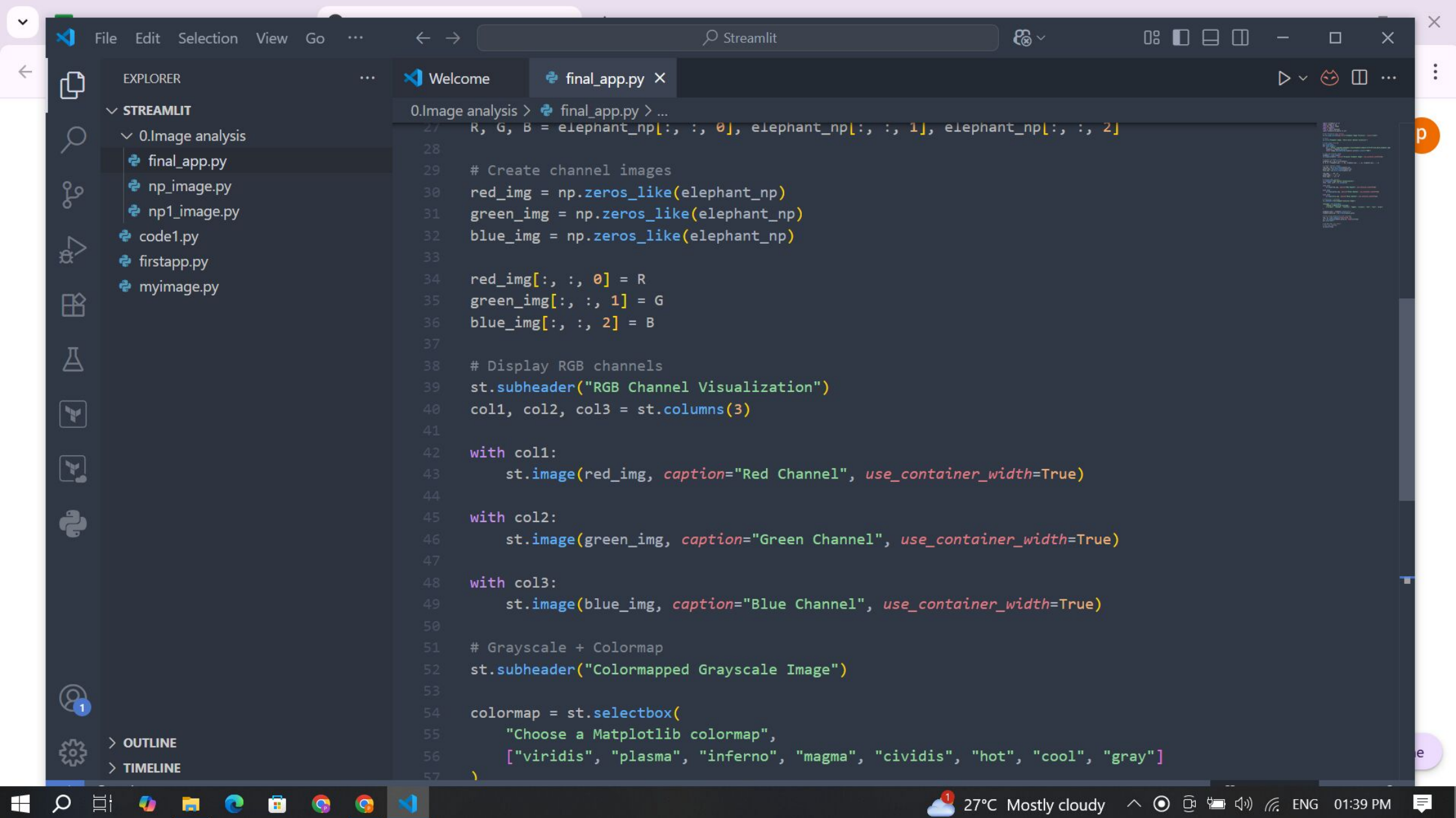
final_app.py

0.Image analysis > final_app.py > ...

```
1 import streamlit as st
2 import numpy as np
3 from PIL import Image
4 import requests
5 from io import BytesIO
6 import matplotlib.pyplot as plt
7
8 # Set Streamlit page config
9 st.set_page_config(page_title="Elephant Image Processor", layout="wide")
10
11 # Title
12 st.title("Elephant Image - Multi-Color Channel Visualizer")
13
14 # Load image from URL
15 @st.cache_data
16 def load_image():
17     url = "https://upload.wikimedia.org/wikipedia/commons/3/37/African_Bush_Elephant.jpg"
18     response = requests.get(url)
19     return Image.open(BytesIO(response.content)).convert("RGB")
20
21 # Load and display image
22 elephant = load_image()
23 st.image(elephant, caption="Original Elephant Image", use_container_width=True)
24
25 # Convert to NumPy array
26 elephant_np = np.array(elephant)
27 R, G, B = elephant_np[:, :, 0], elephant_np[:, :, 1], elephant_np[:, :, 2]
28
29 # Create channel images
30 red_img = np.zeros_like(elephant_np)
```

27°C Mostly cloudy

ENG 01:38 PM



Elephant Image - Multi-Color Channel Visualizer





Original Elephant Image

RGB Channel Visualization



Red Channel



Green Channel



Blue Channel

Colormapped Grayscale Image

Choose a Matplotlib colormap

viridis

▼

